

Statistical Analysis Report

Analysis Summary Template

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1 Summary

Provide a brief overview of the analysis objectives, key findings, and main conclusions. This section should be concise and accessible to non-technical stakeholders. Max 1 paragraph, no bullet points

2 Introduction

Briefly describe the context and motivation for the analysis. What problem are you trying to solve or what question are you trying to answer? List the main objectives of the analysis. Max 1 paragraph, no bullet points.

3 Data

Describe where the data comes from, how it was collected, and any relevant details about data provenance. Max 1 paragraph (usually a few sentences unless the experimental design is described in detail), no bullet points.

```
# Load your data
# data <- read.csv("your_data.csv")

# For demonstration, using built-in dataset
data <- mtcars

# Display structure of the data
str(data)
```

```
'data.frame':  32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num   6  6  4  6  8  6  8  4  4  6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num   3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num   2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num   0  0  1  1  0  1  0  1  1  1 ...
 $ am  : num   1  1  1  0  0  0  0  0  0  0 ...
 $ gear: num   4  4  4  3  3  3  3  4  4  4 ...
 $ carb: num   4  4  1  1  2  1  4  2  2  4 ...
```

```
# Display first few rows
head(data) %>%
  kable(caption = "Dataset sample")
```

Table 1: Dataset sample

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Hornet	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Sportabout											
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

4 Statistical Analysis

Describe the statistical methods and techniques used in this analysis. Max 1 paragraph (usually a few sentences), no bullet points.

5 Results

Present the results of the analysis in a clear and concise manner. Usually there should be a brief summary of each statistical test with an appropriate accompanying graph or table. For a multivariate model with multiple predictors, present the results in a table with the predictor names and the corresponding p-values, accompanied by a graph of the relationship between each predictor and the response variable.

6 References

List any references or citations used in the analysis.

7 Appendix

Session Information

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)
```

```
Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
```

```
[1] LC_COLLATE=English_United Kingdom.utf8
[2] LC_CTYPE=English_United Kingdom.utf8
[3] LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United Kingdom.utf8
```

```
time zone: Europe/London
tzcode source: internal
```

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] knitr_1.50      lubridate_1.9.4 forcats_1.0.1  stringr_1.5.1
[5] dplyr_1.1.4     purrr_1.0.4    readr_2.1.5    tidyr_1.3.1
[9] tibble_3.3.0    ggplot2_3.5.2  tidyverse_2.0.0
```

loaded via a namespace (and not attached):

```
[1] gtable_0.3.6      jsonlite_2.0.0    compiler_4.5.1    tidyselect_1.2.1
[5] scales_1.4.0      yaml_2.3.10       fastmap_1.2.0     R6_2.6.1
[9] generics_0.1.4    pillar_1.10.2     RColorBrewer_1.1-3 tzdb_0.5.0
[13] rlang_1.1.6       stringi_1.8.7     xfun_0.52         timechange_0.3.0
[17] cli_3.6.5         withr_3.0.2       magrittr_2.0.3    digest_0.6.37
[21] grid_4.5.1        hms_1.1.3         lifecycle_1.0.4   vctrs_0.6.5
[25] evaluate_1.0.4    glue_1.8.0        farver_2.1.2      rmarkdown_2.29
[29] tools_4.5.1       pkgconfig_2.0.3   htmltools_0.5.8.1
```